

BUSINESS REQUIREMENT DOCUMENT (BRD)

AI/ML-Powered Financial Intelligence & Automated Management Platform

Project Title:	Expense Tracker with Smart Insighter	Date:	September 25, 2026
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Enrollment Number:	[Your Enrollment Number]	Institute:	[Your Institute Name]

1. Introduction

In the contemporary digital financial landscape, personal expense management has transitioned from manual ledgers to automated digital systems. However, most existing applications act merely as passive data stores that require high manual effort and lack predictive capabilities. **Expense Tracker with Smart Insighter** is an enterprise-grade, full-stack financial intelligence solution designed to transform passive bookkeeping into an active, proactive financial management experience. It integrates automated machine learning pipelines with high-speed RESTful services and interactive visual analytics.

2. Project Overview & Tech Stack

The platform utilizes a decoupled micro-architecture where a high-responsiveness React Single Page Application (SPA) communicates with an asynchronous FastAPI backend backed by relational persistence and Scikit-Learn data science models.

Component	Technology	Architecture & Role
Frontend Client	React 18, Vite	Single Page Application (SPA) with modular hooks & reactive state management
Styling & UI	Vanilla CSS, Lucide React	Dark-mode glassmorphic design system with responsive layouts
Backend Server	FastAPI (Python 3.10+)	Asynchronous high-throughput REST API with OpenAPI validation
Database & ORM	SQLite / PostgreSQL, SQLAlchemy	Relational schema persistence, ACID transactions, and indexed querying
Machine Learning	Scikit-Learn, NumPy, Pandas	TF-IDF NLP categorization, Isolation Forest anomaly scanner, Velocity regression
Security & Auth	JWT, Passlib (Bcrypt)	Stateless JSON Web Tokens with encrypted claims and hashed credentials
Deployment	Vercel (Client), Cloud Server (API)	Global CDN edge distribution with SPA routing rewrite rules

3. Business Problem Statement

Contemporary personal finance applications exhibit significant drawbacks:

- **High Manual Friction:** Manual entry and categorization lead to tracking fatigue and inconsistent record-keeping.
- **Absence of Predictive Foresight:** Retrospective graphs only show past mistakes rather than forecasting impending deficits.
- **Undetected Subscription Creep:** Subtle price increases in recurring monthly services frequently go unnoticed.
- **Lack of Intelligent Outlier Flagging:** Fraudulent, duplicate, or abnormal spending spikes are not isolated in real-time.
- **No Administrative Telemetry:** Lack of centralized oversight to monitor user engagement and platform financial health.

4. Business Objectives

- **Automate Data Capture:** Provide instantaneous manual logging and bulk bank statement CSV ingestion.
- **NLP Auto-Classification:** Automatically classify transaction notes with confidence scoring using machine learning.
- **Predictive Month-End Velocity:** Calculate linear spending velocity to project final monthly balances before overspending occurs.
- **Intelligent Anomaly Radar:** Utilize unsupervised Isolation Forest algorithms to pinpoint statistical outliers.
- **Enterprise-Grade Security & Governance:** Provide secure role-based access control (RBAC) and administrative oversight.

5. Functional Requirements

5.1 User Authentication & Authorization

- Secure User Registration and Login with Bcrypt password hashing (minimum 12 salt rounds).
- Stateless JSON Web Token (JWT) issuance with role-based access control (RBAC: User and Admin).

5.2 Transaction & Statement Management

- Complete CRUD capabilities for income and expense records with category, payment mode, and timestamp attributes.
- Multi-format Bank Statement CSV parser with automated column normalization and batch processing.

5.3 Machine Learning & Financial Intelligence

- **NLP Categorizer:** TF-IDF vectorization paired with classification models for high-accuracy transaction tagging.
- **Continuous Online Learning:** Adaptive retraining model incorporating user category corrections.
- **Predictive Velocity Engine:** Time-series regression estimating month-end burn rate and potential budget overrun.
- **Isolation Forest Anomaly Detection:** Unsupervised outlier modeling highlighting irregular expenditures.
- **Subscription Hike Radar:** Recurring billing tracking engine flagging unexpected price increases.

5.4 Budgets, Analytics & Admin Oversight

- Dynamic category budget progress bars with visual status indicators (Safe, Warning, Exceeded).
- Interactive 60-Day Spending Heatmap matrix displaying daily expense concentration.
- Administrator Oversight Portal with system-wide volume, active user directories, and category distributions.

6. Non-Functional Requirements

Category	Requirement Specification
Performance	API response latency < 150ms for CRUD operations; ML inference latency < 400ms under standard loads.
Security	HTTPS transport encryption, Bcrypt credential hashing, JWT expiration, and SQL injection protection via ORM parameterization.
Scalability	Horizontally scalable stateless backend architecture capable of handling multi-tenant concurrent requests.
Reliability	99.9% uptime target with comprehensive client-side error boundaries and asynchronous exception logging.
Usability	Responsive dark-mode interface optimized for desktop, tablet, and mobile displays with intuitive micro-interactions.

7. User Roles & Responsibilities

Role	Access Scope	Core Responsibilities & Capabilities
Standard User	Personal Account	Manage transactions, import statements, configure budgets, view predictive forecasts, analyze anomalies, and monitor financial health score.
Administrator	Platform System	Access Administrator Portal, monitor platform gross volume, inspect active user accounts, toggle user active status, and evaluate category-wide spending distributions.

8. Project Workflow

- Authentication:** Client submits credentials → FastAPI verifies hash → Issues signed JWT token for session state.
- Transaction Intake:** Transaction entered manually or via CSV upload → NLP Categorizer matches text to target category.
- Intelligence Analysis:** Backend runs Isolation Forest anomaly model and time-series velocity projection.
- Visualization:** React client renders interactive charts, progress bars, and heatmap density matrices.
- Admin Oversight:** Admin logs in → System authenticates admin scope → Telemetry portal displays platform aggregate health.

9. Future Enhancements

- **Open Banking & UPI Ingestion:** Direct integration with automated banking APIs and UPI aggregator protocols.
- **LLM Financial Copilot:** Conversational AI interface for contextual natural language financial queries.
- **Multi-Currency & FX Engine:** Automated real-time conversion rates across international currencies.
- **Mobile Native Apps:** Cross-platform mobile clients for iOS and Android powered by React Native.

10. Conclusion

Expense Tracker with Smart Insider represents a comprehensive, intelligent paradigm in modern financial software engineering. By fusing responsive user-centric design with robust asynchronous backend architectures and actionable machine learning intelligence, the application fulfills all functional, analytical, and security criteria outlined in this Business Requirement Document.